In [47]:
```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [49]:
```
df = sns.load_dataset("titanic")
```

In [51]:
```
df.head()
```

Out[51]:

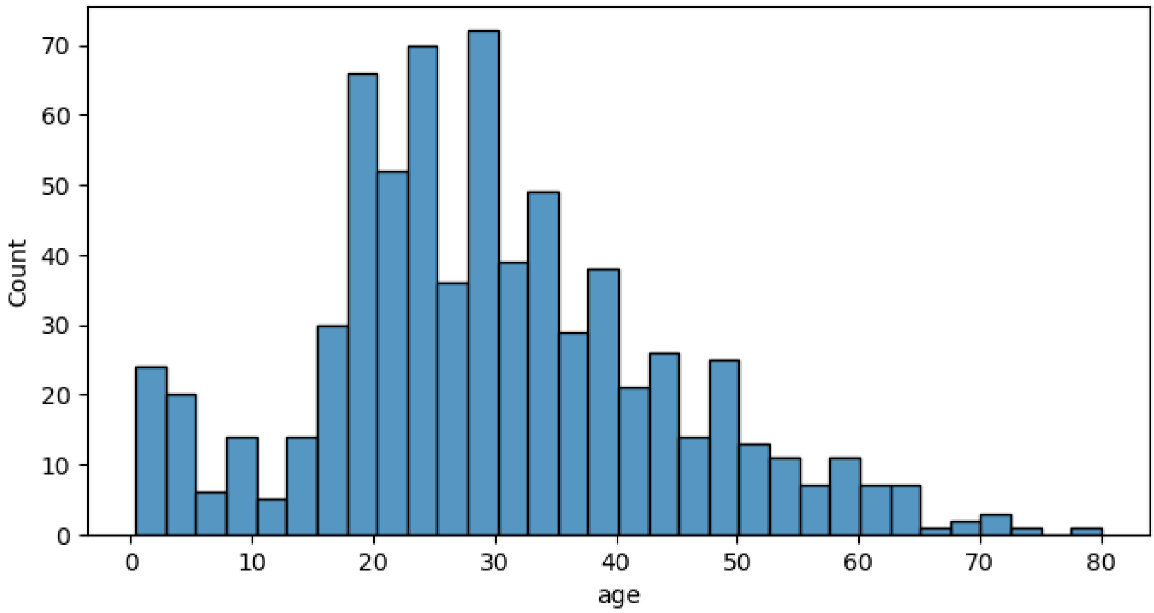
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	ac
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

In [53]:
```
df.describe()
```

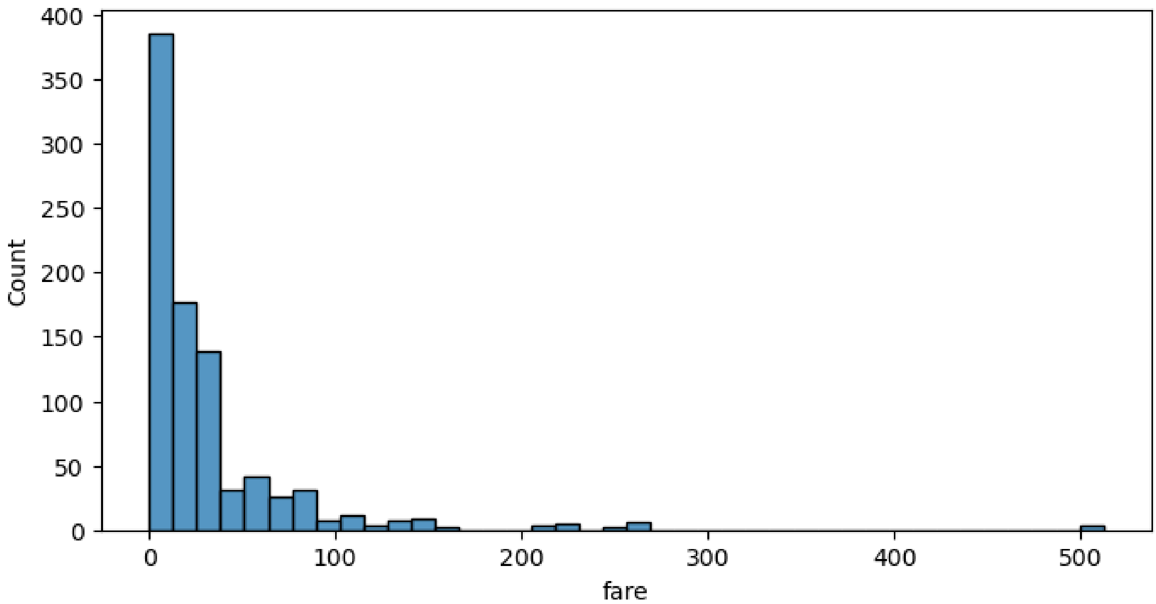
Out[53]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [55]:
```
plt.figure(figsize=(8, 4))
sns.histplot(data=df, x='age', bins=32)
plt.show()
```



In [63]:
```
plt.figure(figsize=(8, 4))
sns.histplot(data=df, x='fare', bins=40)
plt.show()
```



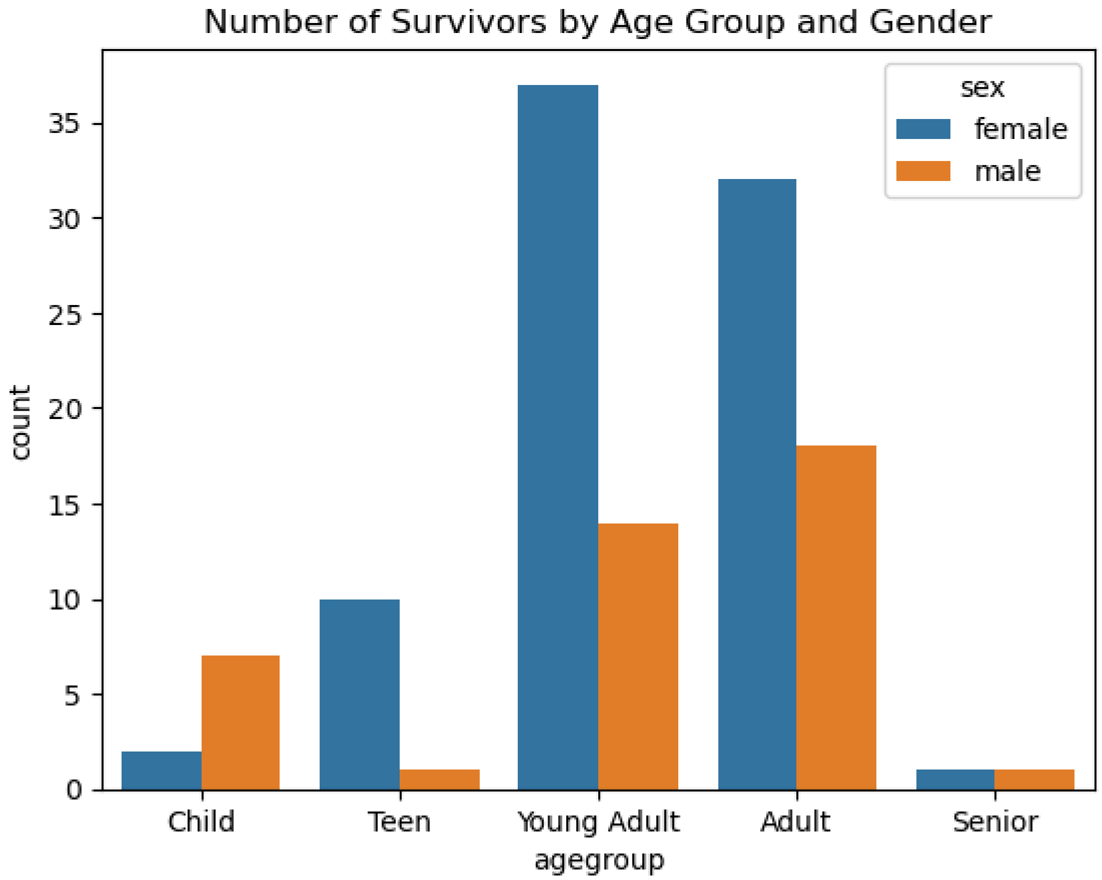
In [85]:
```
df = df.dropna()
```

In [87]:
```
df['agegroup'] = pd.cut(df['age'], bins=[0, 12, 18, 35, 60, 100],
                        labels=['Child', 'Teen', 'Young Adult', 'Adult',
```

In [89]:
```
survivors = df[df['survived'] == 1]
```

In [91]:
```
grouped = survivors.groupby(['agegroup', 'sex'], observed=True).size().re
```

In [93]:
```
sns.barplot(data=grouped, x='agegroup', y='count', hue='sex')
plt.title("Number of Survivors by Age Group and Gender")
plt.show()
```


Number of Survivors by Age Group and Gender
sex
female
male
count
35
30
25
20
15
10
5
0
Child
Teen
Young Adult
Adult
Senior
agegroup

```
In [ ]:
```